

MANDATORY SPARES LIST



Sl. No.	Item	Item Description	Remarks
	PART-A : Mechanical		As per B957-410-80-42-SL-5310 Rev. B
1	PUMP ITEM NO.: 411-PA-CF-101 A/B/C/D/E/F/G/H/I		Not applicable to TTL scope
2	PUMP ITEM NO.: 410-PA-CF-101 A/B/C/D/E/F/G/H/I/J/K/L/M		Not applicable to TTL scope
3	PUMP ITEM NO.: 411-PM-00-108		Not applicable to TTL scope
4	PUMP ITEM NO.: 411-PT-00-109		
A	CENTRIFUGAL PUMPS		
1	Se t of Shaft & Impellers	ONE (1)	Not applicable to TTL scope
2	Se t of Mechanical Seals	ONE (1)	Not applicable to TTL scope
3	Se t of Bearings	ONE (1)	Not applicable to TTL scope
4	IP 66 Rating Oil seals/Bearing isolators	ONE (1)	Not applicable to TTL scope
5	Se t of coupling	ONE (1)	Not applicable to TTL scope
6	Se t of wear rings	ONE (1)	Not applicable to TTL scope
7	Se t of Gaskets	ONE (1)	Not applicable to TTL scope
8	Se t of O-Rings	ONE (1)	Not applicable to TTL scope
B	Positive Displacement Pumps (Rotary)-Lube oil pump		
1	Se t of Rotor assembly	ONE (1)	
2	Se t of Mechanical seals	ONE (1)	
3	Se t of Bearings	ONE (1)	
4	Se t of Packings/gaskets/O-rings	ONE (1)	
C	Steam Turbines (General Purpose)		
1	Dynamically balanced rotor	ONE (1)	Shall be supplied duly dynamically balanced and Mechanical Run Tested, in a permanent metal storage (in vertical condition) containers, with N2 purge.
2	Bearings	ONE (1)	Sleeve Journal / Titling pad thrust bearings
3	Seals	ONE (1)	Brush seals
4	Gaskets/O-rings	ONE (1)	
5	Se t of Laby Seals	ONE (1)	Not applicable to TTL scope
6	Se t of Coupling	ONE (1)	
7	Steam Turbine Casing studs spare Studs set	ONE (1)	
D	GEARBOX		
1	IP 66 Rating Oil seals/Bearing isolators	ONE (1)	

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Sl. No.	Item	Item Description	Remarks
5	PUMP ITEM NO.: 410-PT-00-108		
A	CENTRIFUGAL PUMPS		
1	Set of Shaft & Impellers	ONE (1)	Not applicable to TTL scope
2	Set of Mechanical Seals	ONE (1)	Not applicable to TTL scope
3	Set of Bearings	ONE (1)	Not applicable to TTL scope
4	Set of Gaskets	ONE (1)	Not applicable to TTL scope
5	Set of O-Rings	ONE (1)	Not applicable to TTL scope
6	Set of wear rings	ONE (1)	Not applicable to TTL scope
7	Set of Gaskets	ONE (1)	Not applicable to TTL scope
8	Set of O-Rings	ONE (1)	Not applicable to TTL scope
B	Positive Displacement Pumps (Rotary)-Lube oil pump		
1	Set of Rotor assembly	ONE (1)	
2	Set of Mechanical seals	ONE (1)	
3	Set of Bearings	ONE (1)	
4	Set of Packings/gaskets/O-rings	ONE (1)	
C	Steam Turbines (General Purpose)		
1	Dynamically balanced rotor	ONE (1)	Shall be supplied duly dynamically balanced and Mechanical Run Tested, in a permanent metal storage (in vertical condition) containers, with N2 purge.
2	Bearings	ONE (1)	Sleeve Journal / Titling pad thrust bearings
3	Seals	ONE (1)	Brush seals
4	Gaskets/O-rings	ONE (1)	
5	Set of Laby Seals	ONE (1)	Not applicable to TTL scope
6	Set of Coupling	ONE (1)	
7	Steam Turbine Casing studs spare Studs set	ONE (1)	
D	GEARBOX		
1	IP 66 Rating Oil seals/Bearing isolators	ONE (1)	
6	PUMP ITEM NO.: 410-PM-00-109		Not applicable to TTL scope
	PART-B : ELECTRICAL		As per B957-410-80-42-SL-5310 Rev. B
1	MV induction motors 37 kW & above	One set of spares for each rating & type	Not applicable to TTL scope
2	HV induction motors	One set of spares for each rating & type	Not applicable to TTL scope

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Sl. No.	Item	Item Description	Remarks
PART-C : INSTRUMENTATION			As per B957-410-16-51-SL-5310 Rev-A
1.0	Field Instruments	10% or minimum 01 no. of each type, whichever is higher	Refer to the list below from 1.1 to 1.9
1.1	Pressure Gauges/ Differential Pressure gauge (Normal/Diaphragm type/ Capillary type/Seal type)	10% or minimum 01 no. of each type, range, MOC and rating, whichever is higher	
1.2	Temperature Gauges (Normal as well as Capillary)	10% or minimum 01 no. of each type/ range of temperature gauge, whichever is higher	
1.3	SMART Transmitters for Pressure, Flow (DP), Level(DP),Differential Pressure (DP) and Temperature	10% or minimum 01 no. of each type, range, make and material of construction, whichever is higher	
1.4	Temperature Element: Thermocouple	10% or minimum 01 no. of each type, range, length of Thermocouple with Thermowell, whichever is higher	
1.5	RTDs	10% or minimum 01 no. of each length of RTD with Thermowell, whichever is higher	
1.6	Variable Area Flowmeter (Rotameters)	10% or minimum 01 no. complete set for each type, size, rating and material, whichever is higher	Not applicable to TTL scope
1.7	Turbine Trip SOV	10% or minimum 01 no. of each type, whichever is higher	For pneumatically operated Trunnion mounted ball valve (shutoff valve)
1.8	I/H converter	10% or minimum 01 no. of each type, whichever is higher	
1.9	Instrument items of Purge systems of motors	10% or minimum 01 no. of each type, whichever is higher	Not applicable to TTL scope
2.0	IN-LINE INSTRUMENTS		
2.1	Control Valves	- One set of soft part i.e. Packing, bonnet gasket, stem seal O-ring/Piston O-ring etc. for each valve. - 1 no. spare trim for each type of control valve. - 1 no. actuator diaphragm of each type. - 1 no. actuator spring of each type. - 1 no. of pneumatic accessories (like booster, quick release valve, etc.) of each type.	Pneumatically operated Governing valve
2.2	Shutdown Valves	One set of soft part i.e. Packing, bonnet gasket, stem seal O-ring/Piston O-ring for each valve.	Pneumatically operated Trunnion mounted ball valve (shutoff valve)
2.3	Orifice Plates	10% or minimum 01 no. of blind plates of each size, thickness and material, whichever is higher.	
2.4	Self-Actuating Control Valves	One set of soft part i.e. Packing, bonnet gasket, stem seal O-ring/Piston O-ring etc. for each valve.	

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Sl. No.	Item	Item Description	Remarks
3	System Oriented items		
3.1	Panel mounted instruments	10% or minimum one number of each instrument, whichever is higher.	Applicable for LCS and UCP
3.3.1	Annunciators	a) Power Supply Module 10% subject to minimum 01 No. of each type, whichever is higher	
		b) Tone Generator Card 10% subject minimum 01 no. of each type, whichever is higher	
		c) Flasher Module 10% subject to minimum 01 No. of each type, whichever is higher	
		d) I/O Module 10% subject to minimum 01 No. of each type, whichever is higher	
		e) Lamps 10%.	
		f) Fuses 01 dozen per Annunciator.	
		g) Hooter 10% or minimum one, whichever is higher	
4.0	Other Items		
4.1	Loop powered indicators	10% of Loop powered Indicators used, whichever is higher	Not applicable to TTL scope
4.2	Solenoid Valves	10% or minimum 01 no. of solenoid valves of each type used, whichever is higher	
4.3	Limit Switches (Proximity Type)	10% or minimum 01 no. of Limit Switches of each type used, whichever is higher	
4.4	Air Filter Regulator	10% or minimum 01 no. of Air Filter Regulator of each type used, whichever is higher	
4.5	Smart positioner	10% of installed quantity or minimum 01 nos. of Smart positioner of each type used, whichever is higher	
5.0	Startup and commissioning spares	Startup and Commissioning spares shall be provided as per the vendor's recommendation and the same shall be part of Base Price.	Included in base scope as per tender for TTL scope of supply
6.0	Consumable spares	List of Consumables spares for one year operation shall be provided by the vendor for systems like Analysers, Chromatographs, Machine monitoring systems etc. as applicable for the Package. Consumable spares shall be Price.	Applicable for LCS and UCP